

PARTS LIST

Semi-conductors:

IC1	- LM317 variable voltage regulator
IC2	- 7805, 5V voltage regulator
IC3	- 74HC14 hex Schmitt trigger inverter
BR1	- 2.5A bridge rectifier
BR2	- 1A bridge rectifier
D1-D3	- 1N4001 rectifier diode
ZD1	- 4.7V zener diode
ZD2	- 7.5V zener diode
T1	- BC547 npn transistor
LED1	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

R1	- 220-ohm, 2W
R2, R3, R5	- 1-kilo-ohm
R4	- 680-ohm
R6	- 470-ohm
R7	- 10-ohm, 2W
R8	- 180-ohm
R9-R12	- 300-ohm
POT1	- 1-kilo-ohm pot

Capacitors:

C1	- 100nF, 630V polystyrene
C2	- 10nF ceramic disk
C3, C5, C7	- 0.33 μ F ceramic disk
C4	- 4700 μ F, 35V electrolytic
C6	- 1nF ceramic disk
C8	- 220 μ F, 16V electrolytic

Miscellaneous:

BATT.1	- 6V chargeable battery
X1	- 230V AC primary to 12V, 2.5A secondary transformer
S1	- On/off switch
F1	- 2.5A fuse
F2	- 1A fuse
CON1	- Two-pin connector for AC mains power supply
CON2-CON5	- Two-pin connector
CON6	- Four-pin connector

It is intended to cut only the very high frequency, not to filter the mains power supply. The values of R2 and R3 can be changed depending on IC3 and parameters of the produced pulses.

CON5 provides power-fail signal when the voltage at test point TP2 falls below approximately 8V. Transistor T1 stops conducting and the control unit is intimidated the same by a high signal at pin 3 of CON5.

The threshold voltage of the power-fail signal is regulated with zener diode ZD2 (7.5V) and resistors R4 and R5. Transistor T1 should preferably be switching type, but most of the npn silicon transistors with a high gain will also work well.

The outputs from pins 2, 3 and 4 of connector CON6 provide different signals with a frequency that is double the frequency of the mains power supply (100 Hz). These signals are active near zero crossings of the mains power supply and can be used for several purposes, such as:

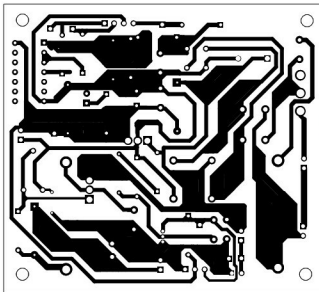


Fig. 2: An actual-size, single-side PCB for the multifunction power supply

